Chapter 3. Probabilistic Regression, Regularization and Kernel Ridge

Linear regression from a probabilistic view, regularization, ridge regression, the kernel trick

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1 Gaussian Likelihood Explains Squared Error

Motivation. Why squared error? If the noise is Gaussian, maximizing the likelihood is exactly minimizing the sum of squares. Each loss in the course corresponds in the same way to a noise model.

Theorem 3.1: MLE under Gaussian noise is least squares

If $y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$ with $\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$, then

$$\log p(\mathbf{y} \mid X, \boldsymbol{\beta}, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2.$$

For fixed σ^2, maximizing the likelihood is minimizing the SSE, so $\hat{\boldsymbol{\beta}}_{\text{MLE}}$ solves the normal equations. Maximizing over σ^2 then gives $\hat{\sigma}^2 = \text{SSE}/n$.

Proof. The density of y_i is $(2\pi\sigma^2)^{-1/2}\exp\{-(y_i - \mathbf{x}_i^\top\boldsymbol{\beta})^2/(2\sigma^2)\}$; independence multiplies densities and the log turns the product into the displayed sum. Differentiating in σ^2: $-\frac{n}{2\sigma^2} + \frac{\text{SSE}}{2\sigma^4} = 0$. ■

2 Ridge Regression

Motivation. Least squares is unstable when $X^\top X$ is nearly singular or there are more features than samples. Adding $\lambda\|\mathbf{w}\|^2$ makes the problem well posed and shrinks the coefficients.

2.1 Definition and closed form

Definition 3.2: ridge

For $\lambda > 0$, $J(\boldsymbol{\beta}) = \frac{1}{2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \frac{\lambda}{2}\|\boldsymbol{\beta}\|^2$.

Theorem 3.3: ridge solution

$\nabla J = X^\top X\boldsymbol{\beta} - X^\top\mathbf{y} + \lambda\boldsymbol{\beta}$ and the unique minimizer is $\hat{\boldsymbol{\beta}}_\lambda = (X^\top X + \lambda\mathbf{I})^{-1}X^\top\mathbf{y}$, even when X is rank deficient.

Proof. $\mathbf{v}^\top(X^\top X + \lambda\mathbf{I})\mathbf{v} = \|X\mathbf{v}\|^2 + \lambda\|\mathbf{v}\|^2 > 0$ for $\mathbf{v} \neq \mathbf{0}$, so the matrix is positive definite, invertible, and J is strictly convex. ■

2.2 MAP interpretation

With prior $\boldsymbol{\beta} \sim N(\mathbf{0}, \tau^2\mathbf{I})$, $-\log p(\boldsymbol{\beta} \mid \mathbf{y}) = \frac{1}{2\sigma^2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \frac{1}{2\tau^2}\|\boldsymbol{\beta}\|^2 + \text{const}$; multiplying by σ^2 gives ridge with $\lambda = \sigma^2/\tau^2$. A tight prior (small τ) means strong shrinkage.

3 Other Penalties and the Lasso

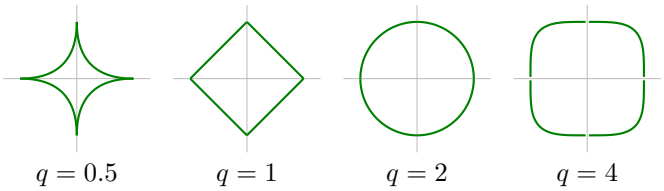
More generally one penalizes $\sum_j |\beta_j|^q$. $q = 2$ is ridge (round level sets, pure shrinkage); $q = 1$ is the lasso, whose diamond-shaped level sets have corners on the axes and produce exact zeros; $0 < q < 1$ is nonconvex.

Theorem 3.4: soft thresholding

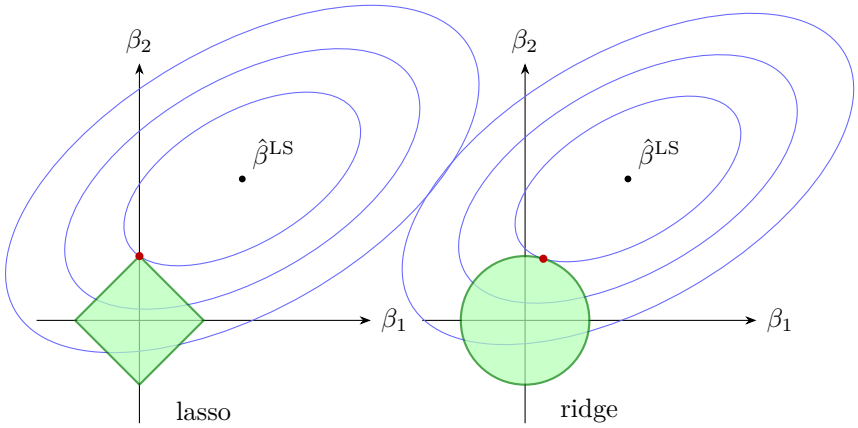
$\arg\min_b \frac{1}{2}(b-z)^2+\lambda|b| = \operatorname{sign}(z)\max(|z|-\lambda,0)$, whereas $\arg\min_b \frac{1}{2}(b-z)^2+\frac{\lambda}{2}b^2 = z/(1+\lambda)$.

Proof. The objective is strictly convex; the optimality condition is $0 \in b - z + \lambda\,\partial|b|$, where $\partial|b| = \{\operatorname{sign} b\}$ for $b \neq 0$ and $[-1,1]$ at 0. If $b > 0$: $b = z - \lambda$, valid iff $z > \lambda$. If $b < 0$: $b = z + \lambda$, valid iff $z < -\lambda$. If $b = 0$: need $z \in \lambda[-1,1]$, i.e. $|z| \le \lambda$. The ridge case is the stationarity condition $b - z + \lambda b = 0$. ■

Lasso moves every coefficient by a fixed amount and cuts small ones to zero (sparsity); ridge rescales proportionally and never produces exact zeros.



Unit balls $\sum_j |\beta_j|^q \le 1$ of the penalties: $q = 1$ is the lasso, $q = 2$ ridge.



Constrained least squares: the smallest RSS contour meets the lasso diamond at a corner (a zero coefficient) and the ridge disk at a generic point.

4 Kernel Ridge Regression

Motivation. Nonlinear regression can be done by mapping inputs to a feature space and fitting ridge there. The solution depends on the data only through inner products, so a kernel lets us work with infinitely many features at the cost of an $n \times n$ linear system.

4.1 Feature-space and sample-space forms

Let $\boldsymbol{\phi}(\mathbf{x}) \in \mathbb{R}^m$ be a feature map and, $\Phi = [\boldsymbol{\phi}(\mathbf{x}_1) \cdots \boldsymbol{\phi}(\mathbf{x}_n)] \in \mathbb{R}^{m\times n}$ (columns are samples). Ridge in feature space minimizes $J = \frac{1}{2}\|\mathbf{y} - \Phi^\top \mathbf{w}\|^2 + \frac{\lambda}{2}\|\mathbf{w}\|^2$, so

$$\nabla J = -\Phi\mathbf{y} + (\lambda \mathbf{I}_m + \Phi\Phi^\top)\mathbf{w}, \qquad \hat{\mathbf{w}} = (\lambda \mathbf{I}_m + \Phi\Phi^\top)^{-1}\Phi\mathbf{y}.$$

Theorem 3.5: push-through identity

$(\lambda \mathbf{I}_m + \Phi\Phi^\top)^{-1}\Phi = \Phi(\lambda \mathbf{I}_n + \Phi^\top\Phi)^{-1}$. Hence with the Gram matrix $K = \Phi^\top\Phi$, $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j) = \boldsymbol{\phi}(\mathbf{x}_i)^\top \boldsymbol{\phi}(\mathbf{x}_j)$,

$$\hat{\mathbf{w}} = \Phi\boldsymbol{\alpha}, \quad \boldsymbol{\alpha} = (K + \lambda \mathbf{I}_n)^{-1}\mathbf{y}, \qquad f(\mathbf{x}) = \boldsymbol{\phi}(\mathbf{x})^\top \hat{\mathbf{w}} = \sum_{i=1}^{n} \alpha_i k(\mathbf{x}_i, \mathbf{x}) = \mathbf{k}_\mathbf{x}^\top (K + \lambda \mathbf{I}_n)^{-1}\mathbf{y}.$$

Proof. $\Phi(\lambda \mathbf{I}_n + \Phi^\top\Phi) = \lambda\Phi + \Phi\Phi^\top\Phi = (\lambda \mathbf{I}_m + \Phi\Phi^\top)\Phi$. Multiply on the left by $(\lambda \mathbf{I}_m + \Phi\Phi^\top)^{-1}$ and on the right by $(\lambda \mathbf{I}_n + \Phi^\top\Phi)^{-1}$ (both positive definite). The two identity matrices have different sizes. ■

Remark 3.6: the kernel trick and cost

$\boldsymbol{\phi}$ appears only through inner products, so only the kernel k is needed. The primal form solves an $m \times m$ system and the dual an $n \times n$ one: the kernel form wins when $m \gg n$ (e.g. the Gaussian kernel, whose feature space is infinite-dimensional). Merely mapping to higher dimensions is feature engineering; computing through k without forming $\boldsymbol{\phi}$ is the kernel trick. An alternative derivation substitutes $\mathbf{w} = \Phi\boldsymbol{\alpha}$ and obtains $K[(K + \lambda \mathbf{I})\boldsymbol{\alpha} - \mathbf{y}] = \mathbf{0}$; when K is singular one may not cancel K, but the displayed $\boldsymbol{\alpha}$ is a solution and all solutions give the same $\hat{\mathbf{w}}$.

4.2 Kernels

- Linear: $k(\mathbf{x}, \mathbf{z}) = \mathbf{x}^\top \mathbf{z}$.
- Quadratic in 2D: $(\mathbf{x}^\top \mathbf{z})^2 = x_1^2 z_1^2 + x_2^2 z_2^2 + 2x_1 x_2 z_1 z_2$, so $\boldsymbol{\phi}(\mathbf{x}) = (x_1^2, x_2^2, \sqrt{2}x_1 x_2)$. In 1D, $(1 + xz)^2$ has $\boldsymbol{\phi}(x) = (1, \sqrt{2}x, x^2)$.
- Gaussian: $k(\mathbf{x}, \mathbf{z}) = \exp(-\|\mathbf{x} - \mathbf{z}\|^2/(2\sigma^2))$; σ is the bandwidth.
- Normalized: $\tilde{k}(\mathbf{x}, \mathbf{z}) = k(\mathbf{x}, \mathbf{z})/\sqrt{k(\mathbf{x}, \mathbf{x})k(\mathbf{z}, \mathbf{z})}$.

Theorem 3.7: Gram matrices are PSD

For any feature-map kernel and any $\mathbf{a} \in \mathbb{R}^n$, $\mathbf{a}^\top K\mathbf{a} = \|\sum_i a_i \boldsymbol{\phi}(\mathbf{x}_i)\|^2 \geq 0$.

Not every similarity function is a valid kernel; positive semidefiniteness of all Gram matrices is the requirement.

4.3 Worked examples

Example 3.8: linear kernel

$\mathbf{x}_1 = (1, 2)$, $\mathbf{x}_2 = (2, 0)$, $\mathbf{y} = (3, 1)$, $\lambda = 1$: $K = \begin{pmatrix} 5 & 2 \\ 2 & 4 \end{pmatrix}$, $(K + \mathbf{I})^{-1} = \frac{1}{26}\begin{pmatrix} 5 & -2 \\ -2 & 6 \end{pmatrix}$, $\boldsymbol{\alpha} = (1/2, 0)$, $\hat{\mathbf{w}} = \frac{1}{2}\mathbf{x}_1 = (1/2, 1)$. At $\mathbf{x} = (1, 0)$: $\mathbf{k}_\mathbf{x} = (1, 2)$ and $f(\mathbf{x}) = 1/2 = \hat{\mathbf{w}}^\top \mathbf{x}$.

Example 3.9: quadratic kernel

$\mathbf{x}_1 = (1,1)$, $\mathbf{x}_2 = (2,0)$, $\mathbf{y} = (3,2)$, $k = (\mathbf{x}^\top \mathbf{z})^2$, $\lambda = 1$. Then $K_{11} = 4$, $K_{12} = 4$, $K_{22} = 16$,

$$(K + \mathbf{I})^{-1} = \frac{1}{69}\begin{pmatrix} 17 & -4 \\ -4 & 5 \end{pmatrix}, \qquad \boldsymbol{\alpha} = \frac{1}{69}\begin{pmatrix} 43 \\ -2 \end{pmatrix}.$$

At $\mathbf{x}_t = (1,0)$, $\mathbf{k}_{\mathbf{x}_t} = (1,4)$ and $f(\mathbf{x}_t) = (43-8)/69 = 35/69 \approx 0.507$. Check in feature space: $\boldsymbol{\phi}(\mathbf{x}_1) = (1,1,\sqrt{2})$, $\boldsymbol{\phi}(\mathbf{x}_2) = (4,0,0)$, $\hat{\mathbf{w}} = \Phi\boldsymbol{\alpha} = (35, 43, 43\sqrt{2})/69$ and $\boldsymbol{\phi}(\mathbf{x}_t) = (1,0,0)$ give the same $35/69$.

5 Selected exercises

Motivation. These exercises make three claims of the chapter precise: ridge is least squares on augmented data with a measurable "number of parameters", the lasso is soft thresholding when the design is orthonormal, and the familiar kernels are valid.

Exercise 3.1 (Ridge as augmented least squares; effective degrees of freedom)

Let $X \in \mathbb{R}^{n\times d}$, $X \neq 0$, $\lambda > 0$, and $J(\boldsymbol{\beta}) = \frac{1}{2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \frac{\lambda}{2}\|\boldsymbol{\beta}\|^2$.

(a) Let $\tilde{X} = \begin{pmatrix} X \\ \sqrt{\lambda}\,\mathbf{I}_d \end{pmatrix}$ and $\tilde{\mathbf{y}} = \begin{pmatrix} \mathbf{y} \\ \mathbf{0} \end{pmatrix}$. Show that $J(\boldsymbol{\beta}) = \frac{1}{2}\|\tilde{\mathbf{y}} - \tilde{X}\boldsymbol{\beta}\|^2$, that $\tilde{X}$ always has full column rank, and recover $\hat{\boldsymbol{\beta}}_\lambda$ from the normal equations of Chapter 1.

(b) The fitted values are $X\hat{\boldsymbol{\beta}}_\lambda = H_\lambda \mathbf{y}$ with $H_\lambda = X(X^\top X + \lambda\mathbf{I})^{-1}X^\top$. Show that $\mathrm{df}(\lambda) := \operatorname{tr} H_\lambda = \sum_{i=1}^{r} \frac{s_i^2}{s_i^2+\lambda}$, where $s_1, \dots, s_r$ are the positive singular values of X.

(c) Show that df is strictly decreasing on $(0,\infty)$, tends to $r = \operatorname{rank} X$ as $\lambda \to 0^+$ and to 0 as $\lambda \to \infty$.

Solution. [Supplement]

(a) The squared norm of a stacked vector is the sum of the squared norms of its blocks, so $\|\tilde{\mathbf{y}} - \tilde{X}\boldsymbol{\beta}\|^2 = \|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \|\mathbf{0} - \sqrt{\lambda}\,\boldsymbol{\beta}\|^2 = 2J(\boldsymbol{\beta})$. If $\tilde{X}\mathbf{v} = \mathbf{0}$, its lower block gives $\sqrt{\lambda}\,\mathbf{v} = \mathbf{0}$, so $\mathbf{v} = \mathbf{0}$: $\tilde{X}$ has full column rank. Since $\tilde{X}^\top\tilde{X} = X^\top X + \lambda\mathbf{I}$ and $\tilde{X}^\top\tilde{\mathbf{y}} = X^\top\mathbf{y}$, the least squares theorem of Chapter 1 gives the unique minimizer $(X^\top X + \lambda\mathbf{I})^{-1}X^\top\mathbf{y} = \hat{\boldsymbol{\beta}}_\lambda$.

(b) Take a full SVD $X = USV^\top$ with $U \in \mathbb{R}^{n\times n}$, $V \in \mathbb{R}^{d\times d}$ orthogonal and $S \in \mathbb{R}^{n\times d}$ having $S_{ii} = s_i$ for $i \le r$ and zeros elsewhere. Then $X^\top X + \lambda\mathbf{I} = V(S^\top S + \lambda\mathbf{I})V^\top$, whose inverse is $V(S^\top S + \lambda\mathbf{I})^{-1}V^\top$, and

$$H_\lambda = US(S^\top S + \lambda\mathbf{I}_d)^{-1}S^\top U^\top.$$

The middle matrix $S(S^\top S + \lambda\mathbf{I}_d)^{-1}S^\top$ is $n \times n$ diagonal with entries $s_i^2/(s_i^2+\lambda)$ for $i \le r$ and 0 otherwise. By cyclicity and $U^\top U = \mathbf{I}$, $\operatorname{tr} H_\lambda$ is the trace of this diagonal matrix, i.e. the sum of these entries.

(c) $r \ge 1$ because $X \neq 0$. Each term has derivative $-s_i^2/(s_i^2+\lambda)^2 < 0$, so df is strictly decreasing. As $\lambda \to 0^+$ each term tends to 1, and as $\lambda \to \infty$ each tends to 0. For full column rank, $\mathrm{df}(0^+) = d = \operatorname{tr} H$ of Exercise 1.1: ridge interpolates continuously between d effective parameters and none. ■

Exercise 3.2 (Orthonormal design: soft, hard and proportional shrinkage)

Assume $X^\top X = \mathbf{I}_d$, let $\lambda > 0$ and $\mathbf{z} = X^\top\mathbf{y}$ (the least squares estimate).

(a) Show that $\frac{1}{2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 = \frac{1}{2}\|\mathbf{z} - \boldsymbol{\beta}\|^2 + c$ with c independent of $\boldsymbol{\beta}$.

(b) Deduce that the lasso $\arg\min_{\boldsymbol{\beta}} \frac{1}{2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \lambda\|\boldsymbol{\beta}\|_1$ is $\hat{\beta}_j = \operatorname{sign}(z_j)\max(|z_j| - \lambda, 0)$, and that ridge with penalty $\frac{\lambda}{2}\|\boldsymbol{\beta}\|^2$ gives $\mathbf{z}/(1+\lambda)$.
(c) Show that best-subset selection $\arg\min_{\boldsymbol{\beta}} \frac{1}{2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \frac{\lambda^2}{2}\|\boldsymbol{\beta}\|_0$, where $\|\boldsymbol{\beta}\|_0$ is the number of nonzero entries, is hard thresholding: $\hat{\beta}_j = z_j$ if $|z_j| > \lambda$ and $\hat{\beta}_j = 0$ if $|z_j| < \lambda$.

Solution. [Supplement]

(a) Using $X^\top X = \mathbf{I}$ and $X^\top \mathbf{y} = \mathbf{z}$,

$$\|\mathbf{y} - X\boldsymbol{\beta}\|^2 = \mathbf{y}^\top\mathbf{y} - 2\boldsymbol{\beta}^\top\mathbf{z} + \boldsymbol{\beta}^\top\boldsymbol{\beta} = \|\mathbf{z} - \boldsymbol{\beta}\|^2 + (\mathbf{y}^\top\mathbf{y} - \mathbf{z}^\top\mathbf{z}).$$

(b) By (a) the lasso objective is $c + \sum_j \left[\frac{1}{2}(z_j - \beta_j)^2 + \lambda|\beta_j|\right]$, a sum of functions of distinct coordinates, so it is minimized by minimizing each term separately. The soft-thresholding theorem gives $\hat{\beta}_j$; the same theorem gives the ridge coordinate $z_j/(1+\lambda)$.

(c) The objective is again $c + \sum_j g_j(\beta_j)$ with $g_j(b) = \frac{1}{2}(z_j - b)^2 + \frac{\lambda^2}{2}\mathbf{1}\{b \neq 0\}$. At $b = 0$, $g_j = z_j^2/2$. Over $b \neq 0$, $g_j(b) \geq \lambda^2/2$, with equality at $b = z_j$ when $z_j \neq 0$. If $|z_j| > \lambda$, then $z_j \neq 0$ and $\lambda^2/2 < z_j^2/2$, so the unique minimizer is $b = z_j$. If $|z_j| < \lambda$, every $b \neq 0$ gives $g_j(b) \geq \lambda^2/2 > z_j^2/2$, so $b = 0$. (At $|z_j| = \lambda$ both values are optimal.) Hard thresholding keeps large coefficients unshrunk, the lasso shifts all survivors by λ, and ridge rescales everything and zeroes nothing. ■

Exercise 3.3 (Building valid kernels)

Call $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ *valid* if every Gram matrix $K = [k(\mathbf{x}_i, \mathbf{x}_j)]_{i,j=1}^n$ is symmetric positive semidefinite.
(a) Show that if k_1, k_2 are valid and $c \geq 0$, then $k_1 + k_2$ and ck_1 are valid, and so is $k(\mathbf{x}, \mathbf{z}) = g(\mathbf{x})g(\mathbf{z})$ for any $g : \mathcal{X} \to \mathbb{R}$.
(b) Show that if $A, B \in \mathbb{R}^{n\times n}$ are symmetric PSD, so is the entrywise product $A \circ B$; deduce that $k_1 k_2$ is valid.
(c) Deduce that $(1 + \mathbf{x}^\top\mathbf{z})^p$ ($p \in \mathbb{N}$) and the Gaussian kernel $\exp(-\|\mathbf{x} - \mathbf{z}\|^2/(2\sigma^2))$ are valid on $\mathbb{R}^m$.

Solution. [Supplement]

(a) Symmetry is clear in each case. For $\mathbf{a} \in \mathbb{R}^n$: $\mathbf{a}^\top(K_1 + K_2)\mathbf{a} = \mathbf{a}^\top K_1\mathbf{a} + \mathbf{a}^\top K_2\mathbf{a} \geq 0$ and $\mathbf{a}^\top(cK_1)\mathbf{a} = c\,\mathbf{a}^\top K_1\mathbf{a} \geq 0$. For $k = g(\mathbf{x})g(\mathbf{z})$, $\mathbf{a}^\top K\mathbf{a} = \sum_{i,j} a_i a_j g(\mathbf{x}_i)g(\mathbf{x}_j) = \left(\sum_i a_i g(\mathbf{x}_i)\right)^2 \geq 0$.

(b) By the spectral theorem $B = \sum_k \mu_k \mathbf{q}_k\mathbf{q}_k^\top$ with $\mu_k \geq 0$; put $\mathbf{b}_k = \sqrt{\mu_k}\,\mathbf{q}_k$, so $B_{ij} = \sum_k b_{k,i}b_{k,j}$. Then

$$\mathbf{a}^\top(A \circ B)\mathbf{a} = \sum_k \sum_{i,j} (a_i b_{k,i}) A_{ij} (a_j b_{k,j}) = \sum_k (\mathbf{a} \circ \mathbf{b}_k)^\top A (\mathbf{a} \circ \mathbf{b}_k) \geq 0.$$

$A \circ B$ is symmetric, and the Gram matrix of $k_1 k_2$ is $K_1 \circ K_2$.

(c) *Step 1 (polynomial kernel).* The linear kernel is valid ($\mathbf{a}^\top K\mathbf{a} = \|\sum_i a_i\mathbf{x}_i\|^2$) and the constant 1 is valid by (a) with $g \equiv 1$. Hence $1 + \mathbf{x}^\top\mathbf{z}$ is valid by (a), and its pth power by (b) and induction on p.

Step 2 (exponential kernel). Let $e(\mathbf{x}, \mathbf{z}) = \exp(\mathbf{x}^\top\mathbf{z}/\sigma^2)$ and $e_N = \sum_{q=0}^N (\mathbf{x}^\top\mathbf{z}/\sigma^2)^q/q!$. By (a), (b) and Step 1, each e_N is valid. For fixed points, the Gram matrices E_N converge entrywise to E, so $\mathbf{a}^\top E\mathbf{a} = \lim_N \mathbf{a}^\top E_N\mathbf{a} \geq 0$.

Step 3 (Gaussian kernel). Expanding $\|\mathbf{x} - \mathbf{z}\|^2 = \|\mathbf{x}\|^2 - 2\mathbf{x}^\top\mathbf{z} + \|\mathbf{z}\|^2$ gives

$$\exp\Big(-\frac{\|\mathbf{x} - \mathbf{z}\|^2}{2\sigma^2}\Big) = g(\mathbf{x})\, e(\mathbf{x}, \mathbf{z})\, g(\mathbf{z}), \qquad g(\mathbf{x}) = \exp\Big(-\frac{\|\mathbf{x}\|^2}{2\sigma^2}\Big),$$

which is valid by (b) applied to e and the kernel $g(\mathbf{x})g(\mathbf{z})$ of (a). ■